

**Department of Electrical and Computer Engineering
North South University**



Project Report

Project Name: Inventory Management System

Semester: Spring 2025

Course: CSE 311L Section: 03

Group: 05

Submitted To:

Faculty: Mr. Md. Ishan Arefin Hossain [IAH]

Group Members

- | | |
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Project Proposal: eCommerce Website

Project Overview

This project aims to develop a fully functional and user-friendly **eCommerce website** using **PHP, MySQL, HTML, CSS(Tailwind CSS), and JavaScript**. This platform will allow users to **browse products, make purchases, and track orders**, while the **admin** will have complete control over **inventory, orders, users, and analytics**.

A personalized product recommendation system will also be integrated using **session tokens or JWT-based authentication**, leveraging user search history to display relevant product suggestions.

Objectives

-  Develop a responsive and intuitive user interface for customers and admins.
 -  Implement secure user authentication and role-based access (customer/admin).
 -  Design a scalable backend system to handle inventory, orders, and analytics.
 -  Enable efficient product browsing, filtering, and recommendations.
 -  Integrate admin features for managing products, orders, and users.
-

System Architecture:

Frontend Technologies :

- **HTML5** (Structure)
- **CSS**
- **Tailwind CSS**
- **JavaScript** (Dynamic interactions, form validation)

Backend Technologies:

- **PHP** (Server-side logic)
- **MySQL** (Database management)

Version Control :

- **Git & GitHub** were used for collaborative development with a proper branching strategy:
 - **main** branch for stable releases
 - **development** a branch for ongoing work
 - Feature branches for individual components
 - Regular commits with meaningful messages

Website Features :

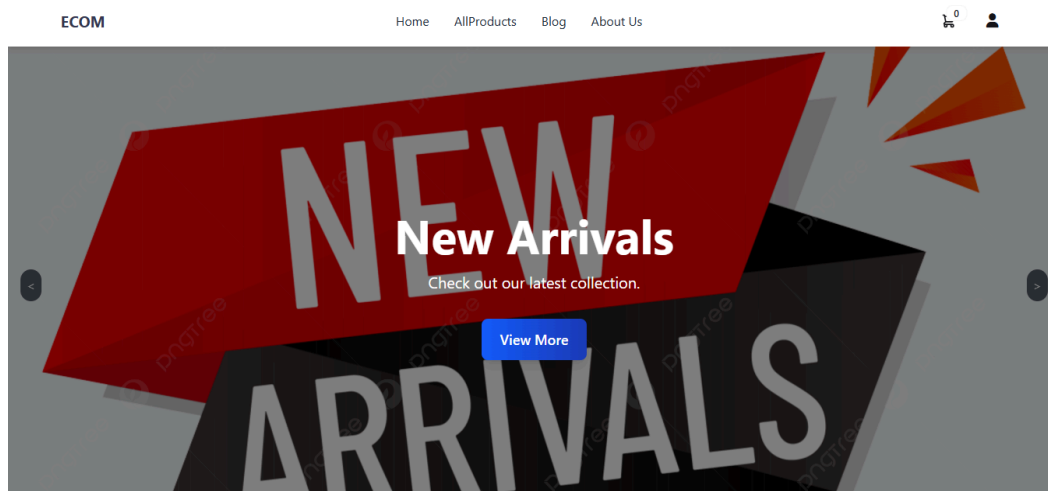
Client-Side (User) Features & Pages

Home Page:

📌 **Overview:** This is the landing page where users first interact with the store.

📌 **Features:**

- ♦ **Hero Banner:** Displays top deals, new arrivals, and company promotions.
- ♦ **Product Categories:** Sections for different product types ().
- ♦ **Top-Selling Products:** Displays the most purchased items.
- ♦ **Discounts & Offers:** Highlights ongoing promotions.

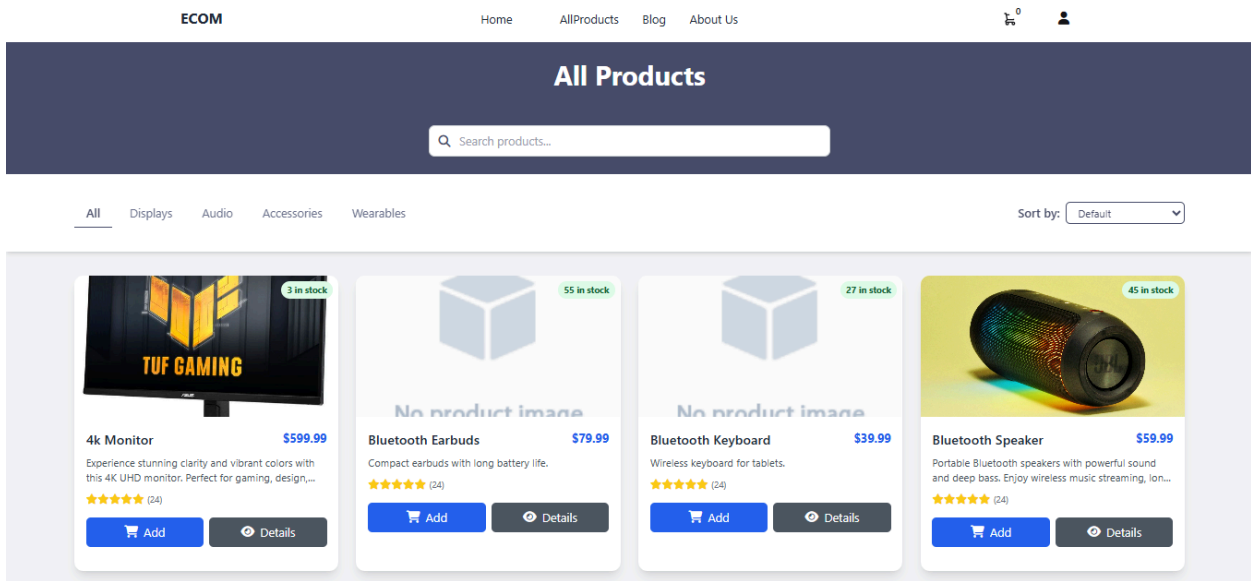


All Product Listing Page :

Overview: This page shows all products in a selected category.

Features:

- ♦ **Filter Options:** Sort by price, popularity, brand, etc.
- ♦ **Search & Auto-Suggestions:** Helps users find products faster.
- ♦ **Pagination:** Loads more products as users scroll.

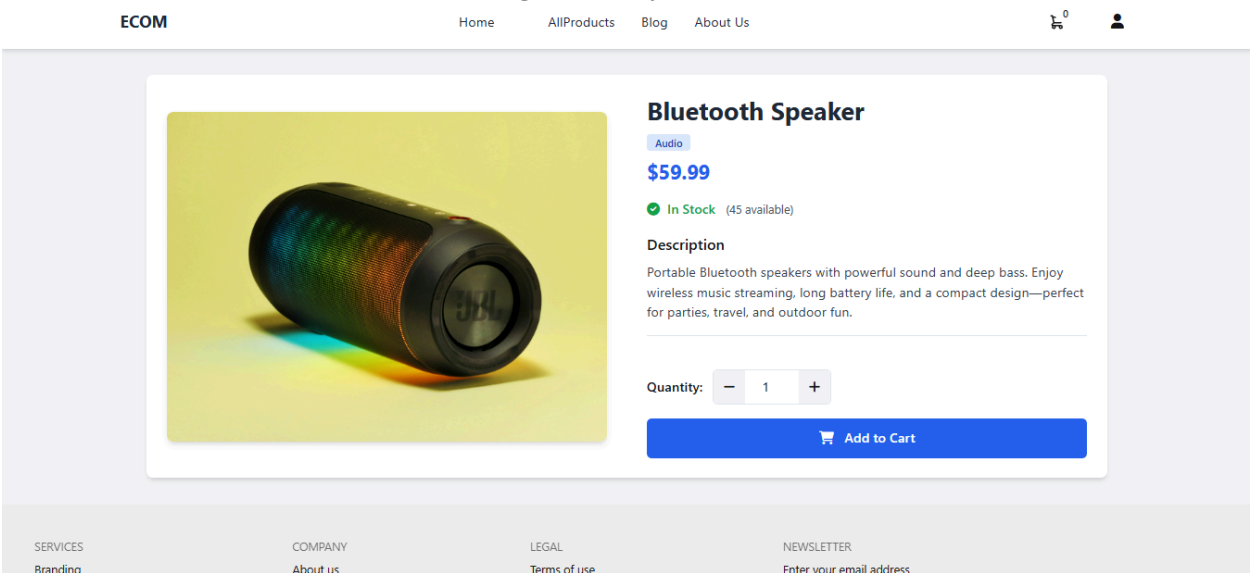


Product Details Page :

Overview: A detailed view of a selected product.

Features:

- ♦ **Product Image Gallery:** Zoom and multiple views.
- ♦ **Product Description & Features:** Full details, specifications, and benefits.
- ♦ **Pricing & Stock Availability:** Real-time updates on stock levels.
- ♦ **"Add to Cart" & "Buy Now" Buttons:** Quick purchase options.
- ♦ **Customer Reviews & Ratings:** Displays user feedback.

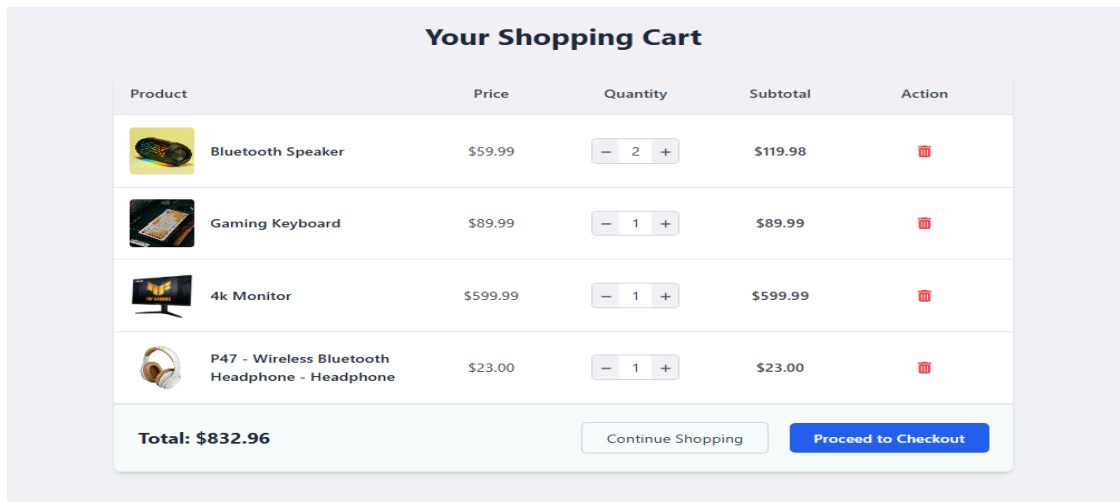


Shopping Cart Page:

📌 **Overview:** Displays all selected items before checkout.

📌 **Features:**

- **Edit Cart Items:** Change quantity or remove items.
- **Subtotal Calculation:** Displays total cost before applying discounts.
- **Proceed to Checkout:** Moves to the payment stage.

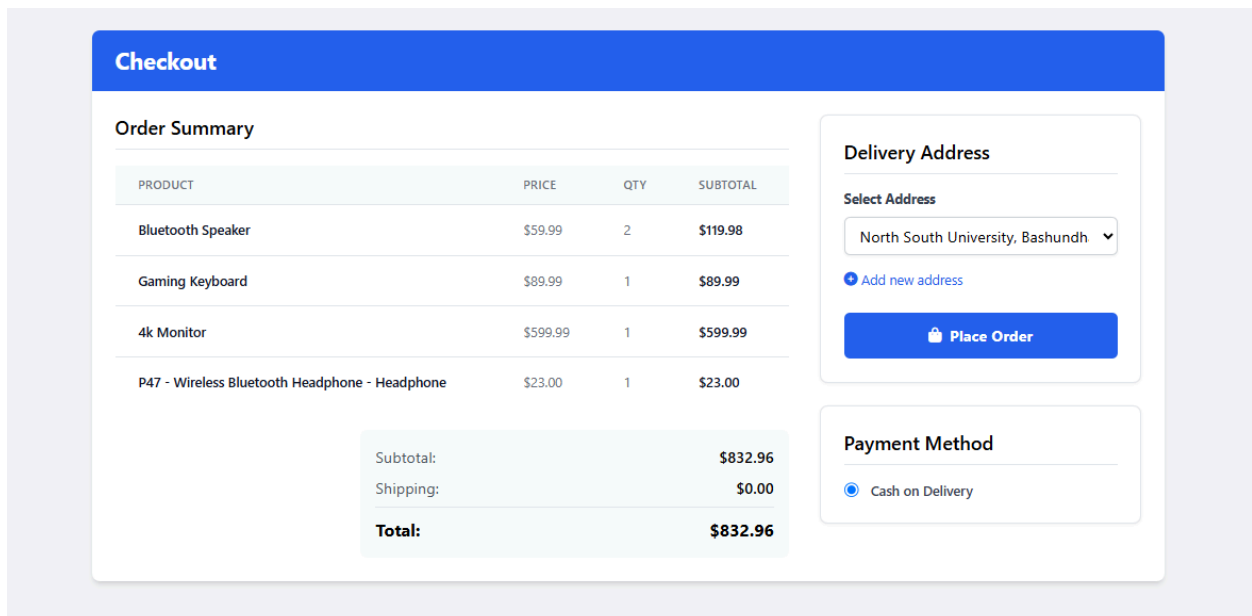


Checkout Page:

📌 **Overview:** Allows users to complete their purchase.

📌 **Features:**

- **Billing & Shipping Address Form:** Saves user details for future purchases.
- **Payment Options:**
- **Order Summary:** Lists all selected products with pricing details.

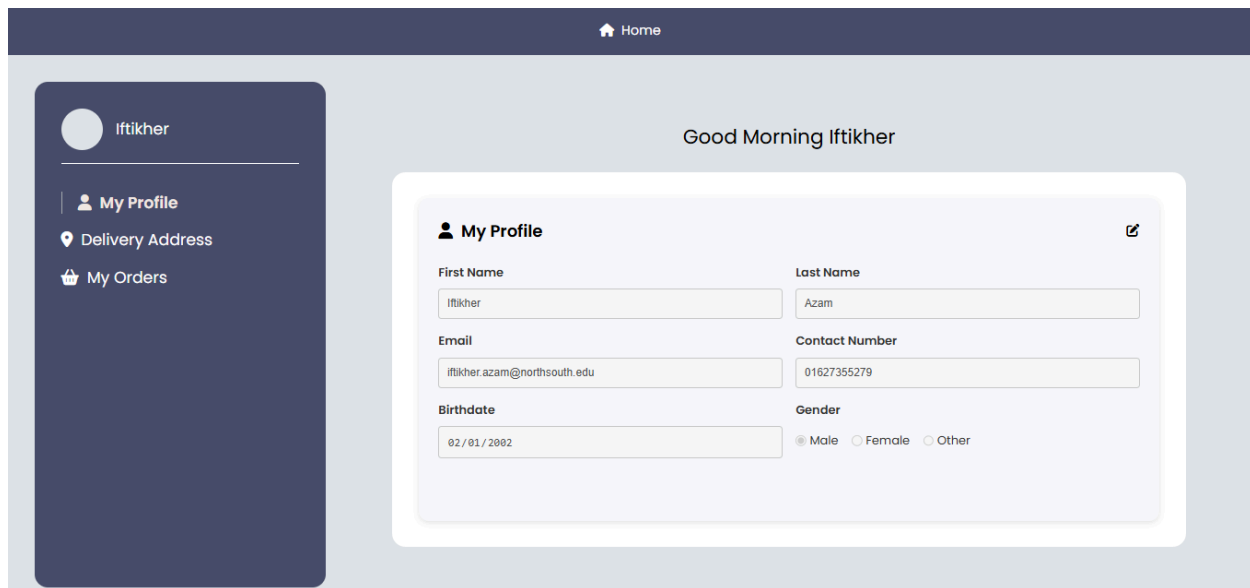


User Account Page:

📌 **Overview:** Users can manage their profiles and track their orders.

📌 **Features:**

- **Update Profile Details:** Edit name, email, address, and password.
- **Order History:** View past purchases and track shipments.



The screenshot shows the 'My Profile' page of a user named Iftikher. The left sidebar contains links for 'My Profile', 'Delivery Address', and 'My Orders'. The main content area displays the user's profile information in a form with fields for First Name, Last Name, Email, Contact Number, Birthdate, and Gender. The user's profile is currently set to Iftikher Azam, with email iftikher.azam@northsouth.edu, contact number 01627355279, birthdate 02/01/2002, and gender Male.

Home

Good Morning Iftikher

My Profile

First Name: Iftikher, Last Name: Azam

Email: iftikher.azam@northsouth.edu, Contact Number: 01627355279

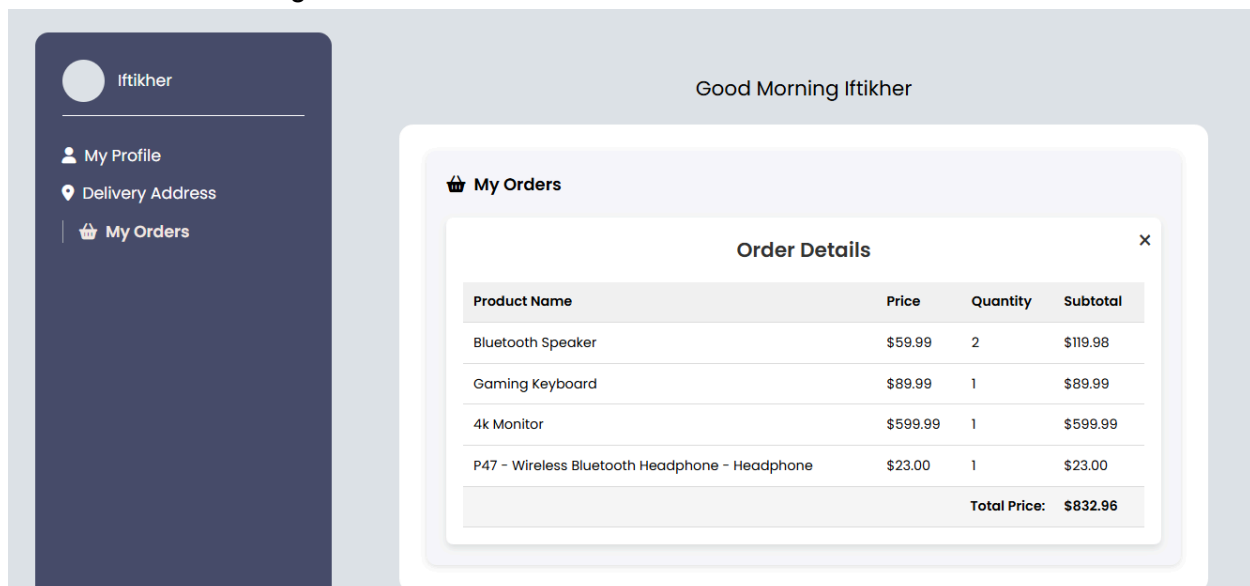
Birthdate: 02/01/2002, Gender: ☒ Male ☐ Female ☐ Other

Order Tracking Page:

📌 **Overview:** Provides real-time order status updates.

📌 **Features:**

- **Track Order with ID:** Users enter their order ID to check the delivery status.
- **Real-Time Updates:** Shows “Processing,” “Shipped,” “Out for Delivery,” and “Delivered” stages.



The screenshot shows the 'My Orders' page of a user named Iftikher. The left sidebar contains links for 'My Profile', 'Delivery Address', and 'My Orders'. The main content area displays a table of order details for a recent purchase. The table lists the product name, price, quantity, and subtotal for each item, along with the total price of \$832.96.

Good Morning Iftikher

My Orders

Order Details

Product Name	Price	Quantity	Subtotal
Bluetooth Speaker	\$59.99	2	\$119.98
Gaming Keyboard	\$89.99	1	\$89.99
4k Monitor	\$599.99	1	\$599.99
P47 - Wireless Bluetooth Headphone - Headphone	\$23.00	1	\$23.00
Total Price:			\$832.96

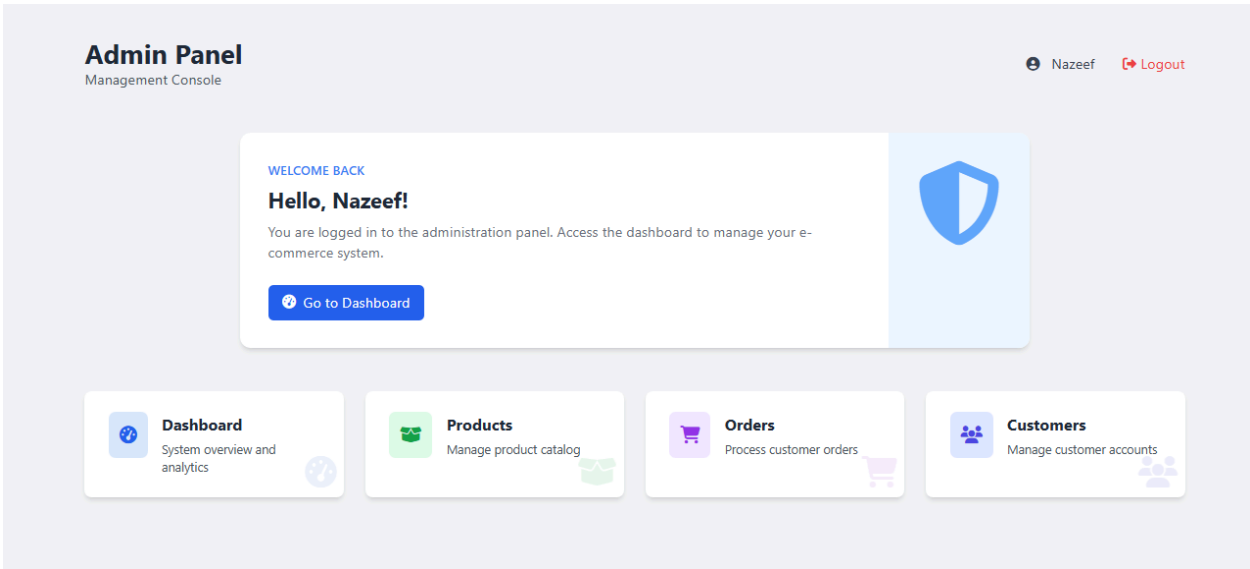
Admin Panel Features & Pages:

Admin Dashboard :

 **Overview:** A control center for managing the eCommerce store.

 **Features:**

-  **Sales Analytics & Revenue Reports.**
-  **Total Orders & Pending Orders.**
-  **Stock Management Alerts.**
-  **Recent Customer Activity.**



AdminPanel

NA

Nazeef

Administrator

Dashboard

Products

Orders

3

Customers

Dashboard Overview

Welcome back, Nazeef!

Here's your store summary for today.

Refresh

Total Revenue

\$3,359.86

Total Orders

8

Pending Orders

3

New Customers

0

Recent Orders

ORDER ID	CUSTOMER	DATE	AMOUNT	STATUS
12	Iftikher Azam	2025-04-21	\$832.96	Pending
11	Iftikher Azam	2025-04-18	\$1199.98	Out For Delivery
10	Iftikher Azam	2025-04-17	\$999.95	Delivered
9	Nazeef Jalal	2025-04-17	\$1469.94	Delivered
8	Nazeef Jalal	2025-04-16	\$889.97	Delivered

View All Orders →

Logout

Product Management Page:

📌 **Overview:** The admin can add, update, and remove products.

📌 **Features:**

- ♦ **Add New Products** (Title, Description, Price, Image Upload).
- ♦ **Edit & Delete Existing Products.**
- ♦ **Update Stock Levels.**

AdminPanel

AdmiAdministrator

Dashboard

Products

Orders

Customers

Logout

Product Management

Product Management

View and manage your product inventory

Search products...

All CategoriesFilter

IMAGE	NAME	CATEGORY	PRICE	STOCK	ACTIONS
	Wireless Headphones	Audio	\$199.99	45	Edit Delete
	Bluetooth Speaker	Audio	\$59.99	43	Edit Delete
	Smart Watch	Wearables	\$149.99	14	Edit Delete
	Gaming Keyboard	Accessories	\$89.99	20	Edit Delete
	4k Monitor	Displays	\$599.99	2	Edit Delete
	USB-C Charger	Accessories	\$29.99	50	Edit Delete

Order Management Page:

📌 **Overview:** Displays all orders placed by customers.

📌 **Features:**

- ♦ **Order ID, Customer Name, Payment Status.**
- ♦ **Change Order Status ("Pending," "Shipped," "Delivered").**
- ♦ **View Order Details & Invoice Generation.**

AdminPanel

NA

Nazeef

Administrator

Dashboard

Products

Orders

Customers

Logout

Order Management

All Orders

8

Pending

3

Processing

1

Delivered

3

Out for Delivery

1

ORDER ID	CUSTOMER	DATE	ITEMS	AMOUNT	STATUS	ACTIONS
#12	Iftikher Azam iftikher.azam@northsouth.edu	Apr 21, 2025	4	\$832.96	Pending	View
#11	Iftikher Azam iftikher.azam@northsouth.edu	Apr 18, 2025	1	\$1199.98	Out For Delivery	View
#10	Iftikher Azam iftikher.azam@northsouth.edu	Apr 17, 2025	5	\$999.95	Delivered	View
#9	Nazeef Jalal nazeef.tawfeeq@gmail.com	Apr 17, 2025	3	\$1469.94	Delivered	View
#8	Nazeef Jalal nazeef.tawfeeq@gmail.com	Apr 16, 2025	3	\$889.97	Delivered	View
#3	Iftikher Azam iftikher.azam@northsouth.edu	Mar 26, 2025	3	\$749.93	Processing	View
#2	Iftikher Azam iftikher.azam@northsouth.edu	Mar 26, 2025	1	\$599.97	Pending	View
#1	Iftikher Azam iftikher.azam@northsouth.edu	Mar 26, 2025	1	\$59.99	Pending	View

Customer Management Page:

📌 **Overview:** Admin can manage registered users.

📌 **Features:**

- **View Customer Profiles & Order History.**

AdminPanel		Customer Management			Nazeef
NA	Nazeef Administrator	Search customers...			
CUSTOMER	EMAIL	GENDER	ORDERS	ACTIONS	
DJ	David Johnson	david@yahoo.com	Other	0	
IA	Iftikher Azam	iftikher.azam@northsouth.edu	Male	6	
NJ	Nazeef Jalal	nazeef.tawfeeq@gmail.com	Male	2	
Showing 1 to 3 of 3 results					< 1 >

Development Plan:

Phase 1: Frontend Development (HTML, CSS, JavaScript) - 3 Weeks

- **Week 1: Project Setup & Home Page Development**
 - Tasks:
 - Set up development environment (code editor, version control - Git).
 - Create project structure (folders for HTML, CSS, JavaScript, images).
 - Design and develop the Home Page layout (HTML).
 - Implement basic styling (Tailwind CSS) for the hero banner, search bar, and product categories.
 - Implement responsive design principles.

- Milestones:
 - Project repository initialized.
 - Basic Home Page structure and styling complete.
- **Week 2: Product Listing & Details Pages**
 - Tasks:
 - Develop the "All Product Listing" page with filter options and pagination.
 - Create the "Product Details" page with image gallery, descriptions, and pricing.
 - Implement basic JavaScript functionality for interactive elements (e.g., image zoom, filter toggles).
 - Begin development of the shopping cart page.
 - Milestones:
 - Product listing and details pages functional.
 - Basic shopping cart page structure.
- **Week 3: Shopping Cart, Checkout, and User Account**
 - Tasks:
 - Complete the "Shopping Cart" page with edit functionality and subtotal calculations.
 - Develop the "Checkout" page with billing/shipping forms and order summary.
 - Create the "User Account" page with profile management and order history.
 - Implement basic form validation using Javascript.
 - Milestones:
 - Shopping cart and checkout processes functional.
 - User account management features implemented.

Phase 2: Backend (PHP, MySQL, Admin Panel) - 2 Weeks

- **Week 4:** Database Setup & Basic Backend Functionality
 - Tasks:
 - Design and create the MySQL database schema (products, users, orders, categories, etc.).
 - Set up PHP environment and establish database connections.
 - Develop backend logic for user registration and login.
 - Begin development of the admin panel structure.
 - Milestones:
 - Database structure complete.
 - User authentication implemented.
 - Admin panel basic structure.
- **Week 5:** Admin Panel Development
 - Tasks:
 - Develop the "Product Management" page (add, edit, delete products).
 - Implement "Order Management" (view orders, update status).
 - Create "Customer Management" (view profiles, order history).

Conclusion

This Inventory Management System and eCommerce platform is designed to mimic a real-world online store with a robust set of features for both users and administrators. Through this project, we aim to gain real experience in full-stack development, database design, user authentication, and dynamic frontend development. Upon completion, this project will showcase our ability to build scalable, responsive, and secure web applications from scratch.